

CASE STUDY: How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?

Introduction

This case study is created for the final project of Google Data Analytics Professional Certificate Course Capstone Project. My role in this case study is a junior data analyst working in the marketing analyst team at *Cyclistic*, a bike-share company based in Chicago, USA. The director of marketing believes that the company's future success depends on maximizing the number of annual memberships. Therefore, our team wants to understand how casual riders and annual members use Cyclistic bikes differently. From these insights, our team will design a new marketing strategy to convert casual riders into annual members.

Therefore, the goal of my team is to design a new marketing strategy to convert casual riders into annual members.

To do that, we need to better understand:

1. How annual members and casual riders differ
2. Why casual riders would buy a membership
3. How digital media could affect their marketing tactics.

The first question, **how annual members and casual riders** differ will be handled in this case study.

In order to perform this study, I used the steps of data analysis process: **Ask, Prepare, Process, Analyze, Share, and Act.**

Tools used in this analysis:

- Data Cleaning and Analysis: Microsoft SQL Server Management Studio
- Visualization: Tableau Public

Step 1: Ask

Background

In 2016, Cyclistic launched a successful bike-share offering. Since then the program has grown to a fleet of 5,824 bicycles that are geotracked and locked into a network of 692 stations across Chicago. The bikes can be unlocked from one station and returned to any other station in the system anytime. Single-ride passes, full-day passes, and annual memberships are the three adjustable price options offered by Cyclistic. Casual riders are consumers who buy single-ride or full-day passes. Annual members are clients who buy yearly memberships.

Business Task

Analyze the differences in how annual members and casual riders use cyclistic bikes. The following will be confirmed in order to do that:

- i. Percentage of casual rider's vs annual members
- ii. Frequency of rides by bike type
- iii. Frequency of rides by month
- iv. Number of rides by day
- v. Number of rides by hour
- vi. Average ride duration by day (in minutes)
- vii. Average ride duration by month (in minutes)

Step 2: Prepare

The Data

Motivate International Inc. has granted this license for the dataset. Anyone can utilize this publicly available data to investigate how various customer kinds are utilizing cyclistic bikes. The data is organized in monthly csv files, the most recent twelve months of data (January, 2025 – December, 2025) were used for this project.

Name	Date modified	Type	Size
 202501-divvy-tripdata.csv	27/07/2026 12:43 pm	Microsoft Excel C...	27,973 KB
 202502-divvy-tripdata.csv	27/07/2026 12:43 pm	Microsoft Excel C...	30,540 KB
 202503-divvy-tripdata.csv	27/07/2026 6:43 pm	Microsoft Excel C...	59,472 KB
 202504-divvy-tripdata.csv	27/07/2026 6:44 pm	Microsoft Excel C...	73,877 KB
 202505-divvy-tripdata.csv	27/07/2026 6:44 pm	Microsoft Excel C...	99,176 KB
 202506-divvy-tripdata.csv	27/07/2026 6:44 pm	Microsoft Excel C...	132,770 KB
 202507-divvy-tripdata.csv	27/07/2026 6:44 pm	Microsoft Excel C...	148,763 KB
 202508-divvy-tripdata.csv	27/07/2026 6:45 pm	Microsoft Excel C...	153,855 KB
 202509-divvy-tripdata.csv	27/07/2026 6:45 pm	Microsoft Excel C...	139,314 KB
 202510-divvy-tripdata.csv	27/07/2026 6:45 pm	Microsoft Excel C...	125,353 KB
 202511-divvy-tripdata.csv	27/07/2026 6:45 pm	Microsoft Excel C...	69,444 KB
 202512-divvy-tripdata.csv	27/07/2026 6:46 pm	Microsoft Excel C...	27,638 KB

Table *combined_trips* was created with UNION ALL containing all tables from the 12 months

```

36  SELECT *
37  INTO combined_data
38  FROM
39  (
40      SELECT * FROM January|
41      UNION ALL
42      SELECT * FROM February
43      UNION ALL
44      SELECT * FROM March
45      UNION ALL
46      SELECT * FROM April
47      UNION ALL
48      SELECT * FROM June
49      UNION ALL
50      SELECT * FROM July
51      UNION ALL
52      SELECT * FROM August
53      UNION ALL
54      SELECT * FROM September
55      UNION ALL
56      SELECT * FROM October

```

Each file has the following columns description:

	Column Name	Data Type
►	ride_id	nvarchar(50)
	rideable_type	nvarchar(50)
	started_at	datetime2(7)
	ended_at	datetime2(7)
	start_station_name	text
	start_station_id	text
	end_station_name	text
	end_station_id	text
	start_lat	float
	start_lng	float
	end_lat	float
	end_lng	float
	member_casual	nvarchar(50)

The files have the following number of rows:

	ride_month	total_rows
1	January	136946
2	February	150249
3	March	294017
4	April	366244
5	May	494793
6	June	666509
7	July	748832
8	August	775931
9	September	703147
10	October	635719
11	November	350786
12	December	138044

Step 3: process

Data cleaning in Microsoft SQL Server Management Studio

Each file was cleaned in Microsoft SQL Server Management Studio and then proceeded to data transformation.

- Incomplete values and missing values were checked in all the columns using IS NULL function for missing values and LEN() function to validate the length of incomplete values. The columns start_station_name, start_station_id, end_station_name, and end_station_id have a sizable number of missing values. Each table had those rows removed.

	ride_...	missing_ride_id	missing_rideable_ty...	missing_started_at	missing_ended...	missing_start_stat...	missing_start_sta...	missing_end_stat...	missing_end_stat...	missing_start_lat	missing_start_lng	missing_end_lat	missing_end_lng
1	Janu...	0	0	0	0	22088	22088	22788	22788	0	0	61	61
2	Febr...	0	0	0	0	25161	25161	26060	26060	0	0	66	66
3	March	0	0	0	0	53080	53080	54936	54936	0	0	232	232
4	April	0	0	0	0	68106	68106	69656	69656	0	0	297	297
5	May	0	0	0	0	98517	98517	101393	101393	0	0	589	589
6	June	0	0	0	0	143204	143204	146848	146848	0	0	961	961
7	July	0	0	0	0	164438	164438	169951	169951	0	0	981	981
8	August	0	0	0	0	168286	168286	173467	173467	0	0	707	707
9	Sept...	0	0	0	0	149516	149516	155756	155756	0	0	589	589
10	Octo...	0	0	0	0	138765	138765	145513	145513	0	0	591	591
11	Nov...	0	0	0	0	72774	72774	77390	77390	0	0	342	342
12	Dec...	0	0	0	0	25523	25523	28194	28194	0	0	119	119

- No duplicate values were found in each file using the COUNT () + GROUP BY + HAVING functions.
- The columns rideable_type and member_casual were checked and the bike type were only electric_bike and classic_bike, and the type of membership were only member and casual.
- In order to compare and analyze the volume of rides, a new table called clean_combined_trips was built using the previous table, clean_combined_trips, with new columns for month, day, hour, and duration_minutes. Records with ride durations of **less than one minute** or **greater than 24 hours** were identified as outliers and removed to improve data quality. This was achieved using a DELETE statement that filtered records where ride_length was less than 1 minute or greater than 1,440 minutes.

```

298
299     SELECT
300     *,
301     DATENAME(MONTH, started_at) AS ride_month,
302     DATENAME(WEEKDAY, started_at) AS day_of_the_week,
303     DATEPART(HOUR, started_at) AS ride_hour,
304     DATEDIFF(MINUTE, started_at, ended_at) AS ride_length
305 INTO clean_combined_trips
306 FROM combined_trips
307 WHERE DATEDIFF(MINUTE, started_at, ended_at) >= 1
308 AND DATEDIFF(MINUTE, started_at, ended_at) <= 1440;

```

Step 4: Analyze

The following SQL queries were made in Microsoft SQL Server Management Studio in order to continue the analysis.

1. Total riders by membership

```

17 SELECT
18     member_casual AS membership_type,
19     COUNT(*) AS total_rides_by_member,
20     SUM(COUNT(*)) OVER () AS total_rides,
21     CAST(
22         ROUND(
23             COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (),
24             2
25         ) AS DECIMAL(5,2)
26     ) + '%' AS percentage_membership_type
27 FROM clean_combined_trips
28 GROUP BY member_casual
29 ORDER BY member_casual;

```

2. Frequency of rides by bike type

```

SELECT
    member_casual,
    rideable_type,
    COUNT(*) AS total_rides
FROM clean_combined_trips
GROUP BY |
    member_casual, rideable_type
ORDER BY
    member_casual, rideable_type DESC;

```

3. Ride Volume by Month

```
SELECT
    ride_month,
    member_casual AS membership_type,
    COUNT(*) AS total_rides
FROM clean_combined_trips
GROUP BY
    ride_month, member_casual
ORDER BY
    CASE ride_month
        WHEN 'January' THEN 1
        WHEN 'February' THEN 2
        WHEN 'March' THEN 3
        WHEN 'April' THEN 4
        WHEN 'May' THEN 5
        WHEN 'June' THEN 6
        WHEN 'July' THEN 7
        WHEN 'August' THEN 8
        WHEN 'September' THEN 9
        WHEN 'October' THEN 10
        WHEN 'November' THEN 11
        WHEN 'December' THEN 12
    END;
```

2. Ride Volume by Day

```
SELECT
    day_of_the_week,
    member_casual AS type_membership,
    COUNT(*) AS total_rides
FROM clean_combined_trips
GROUP BY day_of_the_week, member_casual
ORDER BY
    CASE day_of_the_week
        WHEN 'Monday' THEN 1
        WHEN 'Tuesday' THEN 2
        WHEN 'Wednesday' THEN 3
        WHEN 'Thursday' THEN 4
        WHEN 'Friday' THEN 5
        WHEN 'Saturday' THEN 6
        WHEN 'Sunday' THEN 7
    END,
    member_casual;
```

3. Volume of rides by Hour

```

SELECT
    ride_hour,
    member_casual AS membership_type,
    COUNT(*) AS total_rides
FROM clean_combined_trips
GROUP BY
    ride_hour,
    member_casual
ORDER BY
    ride_hour;

```

4. Average ride duration by day (in minutes)

```

SELECT
    day_of_the_week,
    member_casual AS membership_type,
    AVG(ride_length) AS avg_ride_length
FROM clean_combined_trips
GROUP BY day_of_the_week, member_casual
ORDER BY
    CASE day_of_the_week
        WHEN 'Monday' THEN 1
        WHEN 'Tuesday' THEN 2
        WHEN 'Wednesday' THEN 3
        WHEN 'Thursday' THEN 4
        WHEN 'Friday' THEN 5
        WHEN 'Saturday' THEN 6
        WHEN 'Sunday' THEN 7
    END;

```

5. Average ride duration by month (in minutes)

```

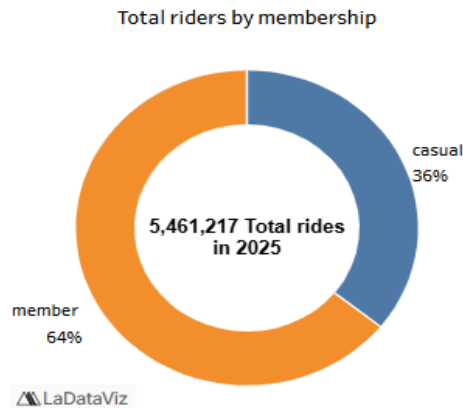
SELECT
    ride_month,
    member_casual AS membership_type,
    AVG(CAST(ride_length AS FLOAT)) AS average_ride_duration_minutes
FROM clean_combined_trips
GROUP BY
    ride_month,
    member_casual
ORDER BY
    CASE ride_month
        WHEN 'January' THEN 1
        WHEN 'February' THEN 2
        WHEN 'March' THEN 3
        WHEN 'April' THEN 4
        WHEN 'May' THEN 5
        WHEN 'June' THEN 6
        WHEN 'July' THEN 7
        WHEN 'August' THEN 8
        WHEN 'September' THEN 9
        WHEN 'October' THEN 10
        WHEN 'November' THEN 11
        WHEN 'December' THEN 12
    END,
    member_casual;

```

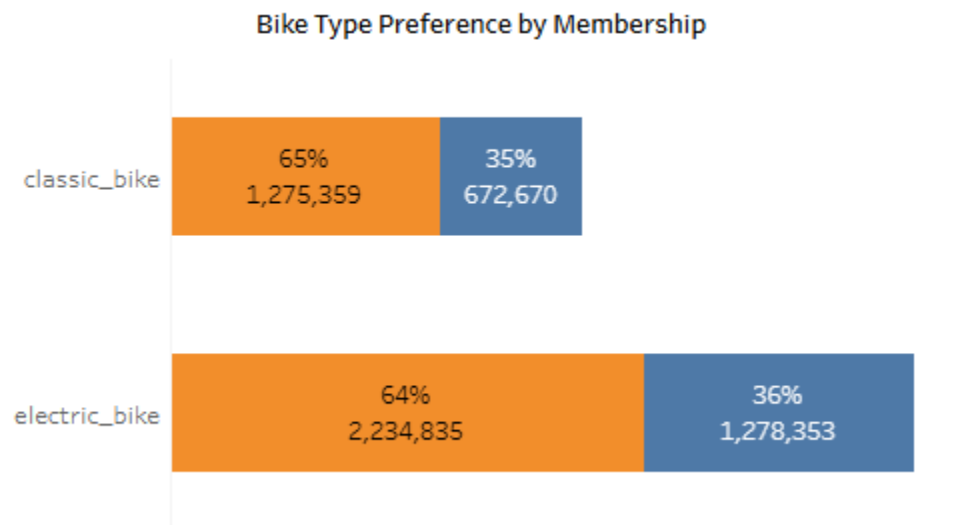

Step 5: Share

Visualizations created in Tableau

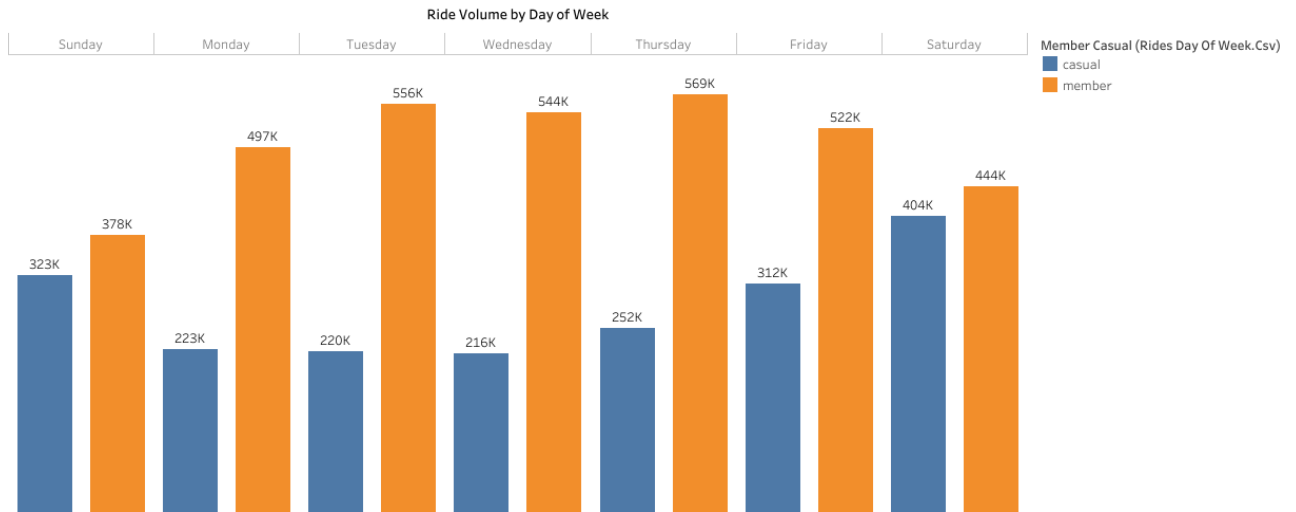
1. Total riders by membership: In 2025 annual members took the most rides (64%) compared to casual riders (36%) from a total of 5,461,217 rides



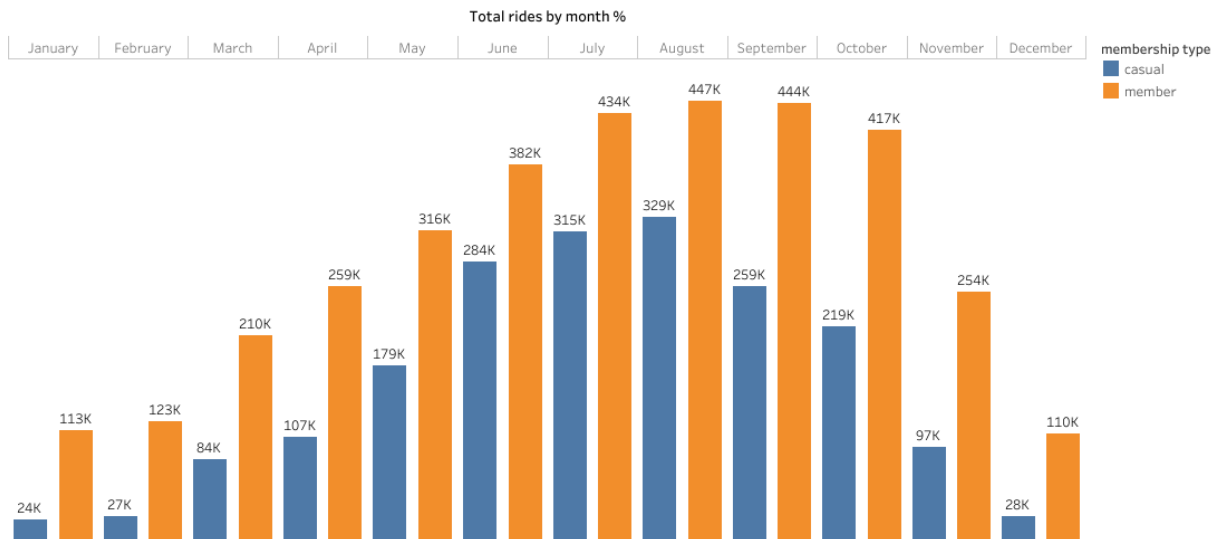
2. **Frequency of rides by bike type:** The breakdown of bike types reveals that compared to casual riders, a greater proportion of annual members ride classic bikes. Additionally, annual members utilize electric bikes more frequently than casual riders.



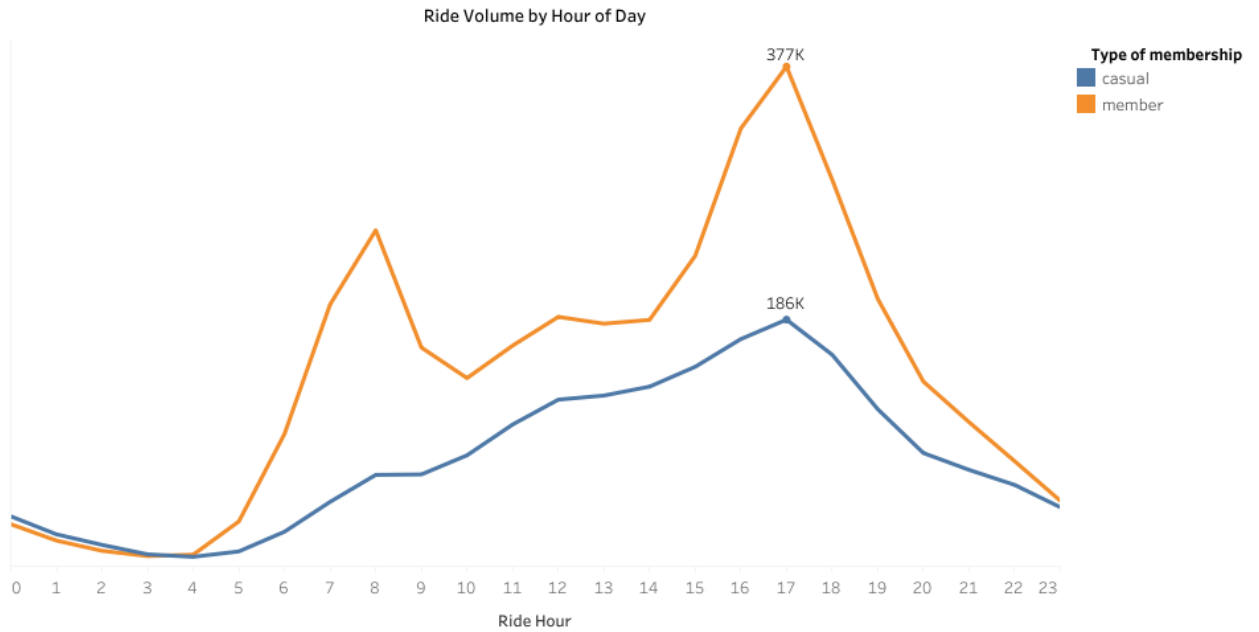
3. **Ride Volume by Day of the Week:** This graphic shows that weekday rides are more common among annual members, whereas weekend bike rentals are more common among casual riders.



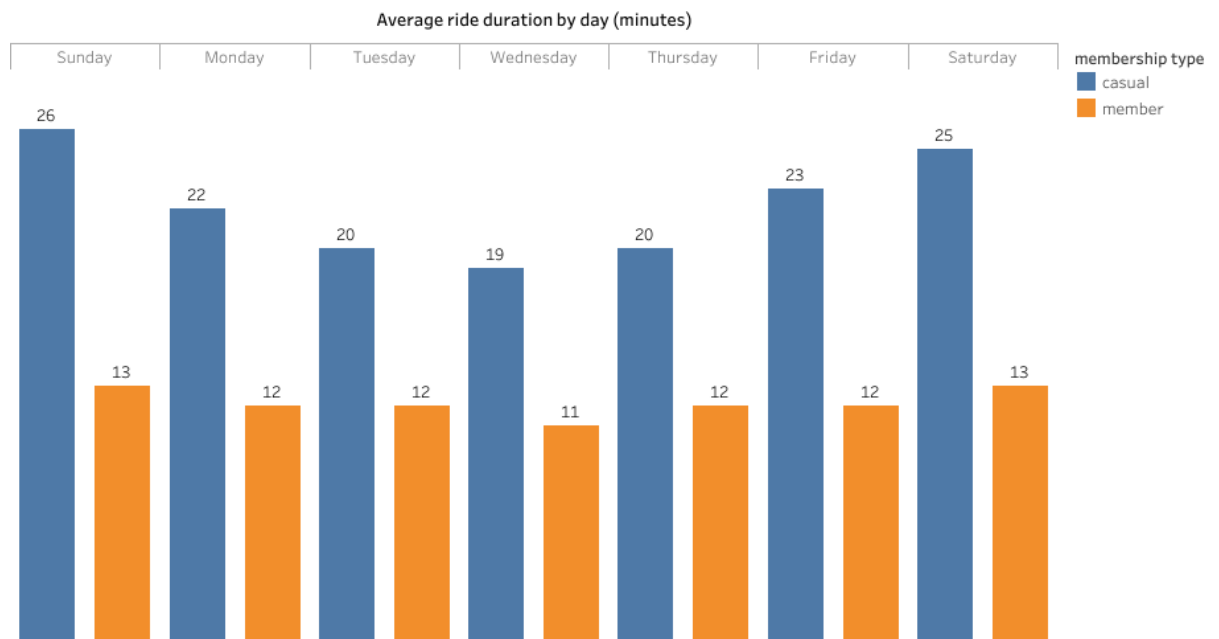
4. **Ride volume of the month:** It is reasonable to draw the conclusion that bike rent increases on hot and sunny days, beginning in the spring (March to May) and increasing for both annual and casual members during the summer (June to August).



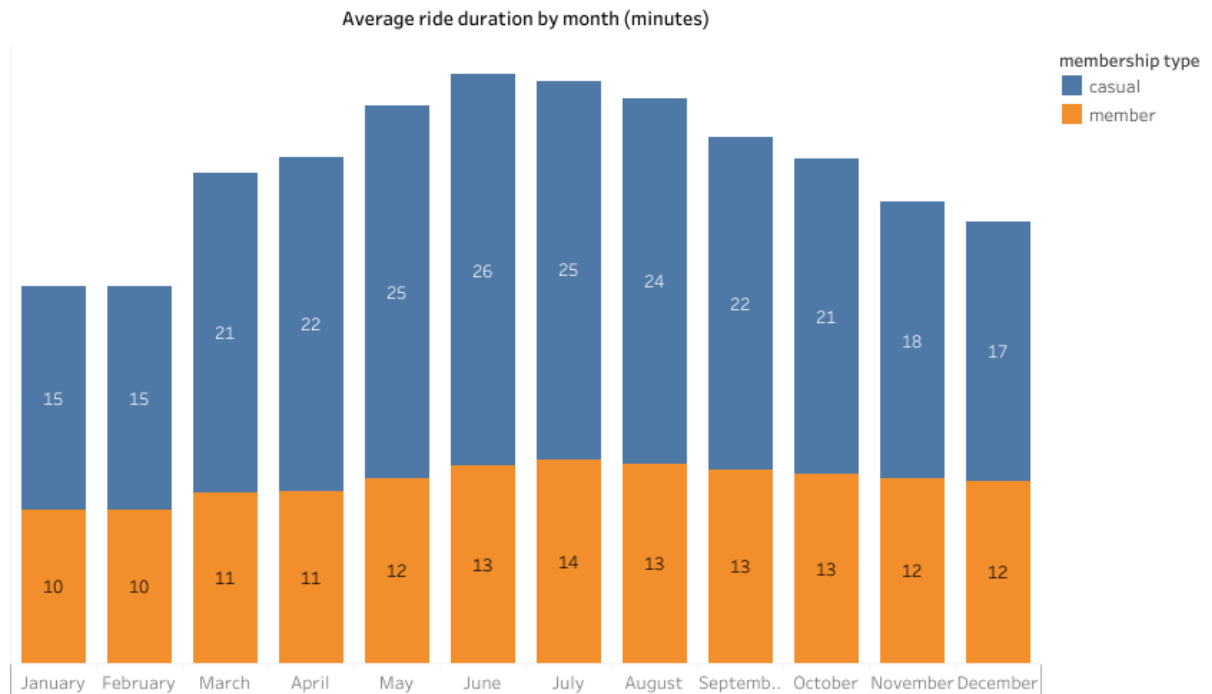
5. **Volume of rides by hour:** This graph shows that while casual riders rent bikes gradually from 6 AM to 5 PM, annual members rent bikes at their highest rate during rush hours (morning 6 to 10 AM and afternoon 3 to 7 PM).



6. **Average ride duration by day (in minutes):** The average ride duration of casual riders is significantly longer than that of annual members, with a peak on Saturday and Sunday, as shown in this graph.



7. **Average ride duration by month (in minutes):** This graph shows that, even when split down by month over the entire year of 2025, casual riders' average ride duration is still significantly longer than that of yearly members.



Step 6: Act

Takeaways

- Annual members completed **64%** of all rides in 2025 (3.49 **million** rides) while casual riders accounted for **36%** (1.97 **million** rides). This indicates that annual members are the primary users of the Cyclistic bike-sharing service.
- Casual rides averaged **20–26 minutes** while annual member rides averaged **11–14 minutes**, this suggests that casual riders mainly use bikes for leisure, sightseeing or recreational activities whereas members use bikes for shorter, purpose-driven trips.
- Annual members completed **2.23 million** electric-bike rides compared with **1.28 million** by casual riders. This indicates strong demand for electric bikes regardless of membership type.

Recommendations

- Target casual riders with membership promotions on weekends to convert casual riders into annual members.
- Launch seasonal marketing campaigns before summer (April–May) to attract riders ahead of the peak riding season.
- Highlight membership savings by showing casual riders how much they could save with an annual membership.
- Increase the availability of electric bikes during peak months and at high-demand locations to meet rider demand.

Tableau Dashboard

